



Cambridge International AS & A Level

PHYSICS

9702/33

Paper 3 Advanced Practical Skills 1

October/November 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	Value of I with unit and in the range 20–120 mA.	1
1(b)	Six (or more) I values with different resistance pairings of R_1 , R_2 without help from the Supervisor scores 5 marks, five sets scores 4 marks, etc.	5
	Range: 33 Ω and 47 Ω resistors used as a pair and 68 Ω and 82 Ω resistors used as a pair.	1
	Column headings: Each column heading must contain a quantity and a unit. The presentation of quantity and unit must conform to accepted scientific convention, e.g. $1/I$ / A ⁻¹ , $\left(\frac{R_1 R_2}{R_1 + R_2}\right)$ / Ω .	1
	Consistency: <u>All</u> values of I must be given to the nearest 0.1 mA or <u>all</u> to the nearest 0.01 mA.	1
	Significant figures: All values of $\left(\frac{R_1 R_2}{R_1 + R_2}\right)$ must be given to 2 s.f. or 3 s.f.	1
	Calculation: Values of $1/I$ and $\left(\frac{R_1 R_2}{R_1 + R_2}\right)$ are correct.	1

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Question	Answer	Marks
1(c)(i)	Axes: Sensible scales must be used, no awkward scales (e.g. 3:10 or fractions). Scales must be chosen so that the plotted points occupy at least half the graph grid in both the x and y directions. Axes must be labelled with the quantity that is being plotted. Scale markings should be no more than three large squares apart.	1
	Plotting of points: All observations in the table must be plotted on the grid. Diameter of plotted points must be $\leq$ half a small square. Points must be plotted to an accuracy of half a small square in both x and y directions.	1
	Quality: All points in the table (at least 5) must be plotted on the grid. Trend of points on graph must be correct. It must be possible to draw a straight line that is within $\pm 1 \Omega$ on the $\left(\frac{R_1 R_2}{R_1 + R_2} \right)$ axis of all plotted points.	1
1(c)(ii)	Line of best fit: Judge by the balance of all points on the grid about the candidate's line (at least 5 points). There must be an even distribution of points either side of the line along the full length. Allow one anomalous point only if clearly indicated (i.e. circled or labelled) by the candidate. There must be at least five points left after the anomalous point is disregarded. Line must not be kinked or thicker than half a small square.	1

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Question	Answer	Marks
1(c)(iii)	Gradient: The hypotenuse of the triangle used must be greater than half the length of the drawn line. Both read-offs must be accurate to half a small square in both the x and y directions. Method of calculation must be correct, e.g. $\Delta y / \Delta x$. Gradient sign on answer line matches graph drawn.	1
	y-intercept: Correct read-off from a point on the line and substituted into $y = mx + c$ or an equivalent expression. Read-off must be accurate to half a small square in both x and y directions. or Intercept read directly from the graph at $\left(\frac{R_1 R_2}{R_1 + R_2} \right) = 0$, accurate to half a small square.	1
1(d)(i)	Value of P = candidate's gradient and value of Q = candidate's y-intercept. Values must not be written as fractions.	1
	Unit for P is correct e.g. $\Omega^{-1}A^{-1}$ or $\Omega^{-1}mA^{-1}$ or V^{-1} and unit for Q is correct e.g. A^{-1} or mA^{-1}.	1
1(d)(ii)	E and Z correctly calculated from P and Q using: $E = 1 / P$ and $Z = EQ$ or $Z = Q / P$ and units for E (V) and Z (Ω) correct.	1

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Question	Answer	Marks
2(a)(i)	$d = 11.0 \pm 0.5 \text{ cm.}$	1
	Raw value(s) of d recorded to the nearest millimetre.	1
2(a)(ii)	A calculated correctly.	1
2(a)(iii)	Number of significant figures in A is linked to the number of significant figures in d .	1
2(b)(i)	Value of t in the range $0.5 \leq t \leq 1.0 \text{ s}$ and given to 0.1 s or better.	1
	Repeated measurement of t .	1
2(b)(ii)	Percentage uncertainty based on an absolute uncertainty in the range 0.2–0.5 s. If repeated readings have been taken, then the uncertainty can be half the range (but not zero) if working is clearly shown. Correct method of calculation to find percentage uncertainty.	1
2(b)(iii)	Mass in the range 2.0–10.0 g and measured to the nearest 0.1 g or better and with unit.	1
2(c)(i)	Second value of d .	1
2(c)(ii)	Second value of t is larger than the first value of t .	1
2(d)(i)	Two values of k calculated correctly. The final values must not be written as fractions.	1
2(d)(ii)	Valid comment consistent with the calculated values of k , testing against a criterion stated by the candidate.	1

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Question	Answer	Marks
2(e)(i)	A Two readings are not enough to draw a (valid) conclusion (not “not enough for accurate results”, “few readings”). B Difficulty with filter paper at the start, e.g. paper not horizontal/paper too high to view against ruler/parallax in viewing start position/paper held by hand which moves/metre rule not vertical. C Difficult to start stop-watch and release filter paper at the same time. D Problem with fall of filter paper with a reason, e.g. paper does not fall vertically/erratic path/hits boss or stand/misses bench/papers separate while falling. E Difficulty with judging when to stop the stop-watch <u>with a reason</u>, e.g. difficult to align head at bench level/papers arrive separately. F Times are small so large error/uncertainty (in t) or high <u>percentage</u> uncertainty in t. <i>1 mark for each point up to a maximum of 4.</i>	4

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Question	Answer	Marks
2(e)(ii)	A Take more readings <u>and</u> plot a graph or take more readings <u>and</u> compare k values (not “repeat readings” on its own). B Improved method to hold/release filter paper, e.g. horizontal bar level with top of rule or use a set square with detail, e.g. use set square between bench and ruler or use plumb-line. C Video/record/film with timer/frame-by-frame. D Use more filter papers/heavier or thicker paper/glue together or switch off air-conditioning/close windows/use a wind shield. E Use a pressure sensor below/position sensor above or below. F Use a greater distance to fall through or use larger diameters/use fewer papers/use lighter papers/use thinner papers <i>(Credit once only if heavier papers suggested for D and lighter papers suggested for F.)</i> <i>1 mark for each point up to a maximum of 4.</i>	4